

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

DECEMBER EXAMINATION

UCCN1213 FUNDAMENTALS OF COMPUTER AND INFORMATION SECURITY /
UCCN1223 CYBERSECURITY

WEDNESDAY, 21 DECEMBER 2022

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY
BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instruction to Candidates :

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in section B, marks will only be awarded for the FIRST question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

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Section A (Compulsory Questions)

- Q1. (a) The two main types of cryptography are symmetric and asymmetric cryptography. Alice wants to use symmetric cryptography to send a secret message to Bob. Draw a diagram to show how Alice sends the secret message to Bob. (4 marks)
- (b) Suppose Alice and Bob wish to use Rivest-Shamir-Adleman (RSA) encryption as their method to send secret message to each other. Given the prime number $p = 7$ and $q = 9$. Assume the value of $e = 5$, $d = 4$, and the value of message $M = 12$. Calculate and show the steps to encrypt the message. (5 marks)
- (c) Referring to the substitution table shown in Figure 1, given the password 'HONOURS', encrypt the message 'Never Ever Give Up' using Vigenere cipher. (6 marks)

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
A	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
B	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a
C	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b
D	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c
E	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d
F	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e
G	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f
H	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g
I	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h
J	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i
K	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j
L	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k
M	m	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l
N	n	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m
O	o	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n
P	p	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
Q	q	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p
R	r	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q
S	s	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r
T	t	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s
U	u	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t
V	v	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u
W	w	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v
X	x	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
Y	y	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
Z	z	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y

Figure 1

- (d) Raymond has changed his old password to a new password for his email account. Compute the complexity of Raymond's password if his new password is `r@Y35#`. (4 marks)

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Q1. (Continued)

- (e) Biometrics refers to any measure used to uniquely identify a person based on biological or physiological traits. Give **THREE (3)** examples of common biometric identifiers. (6 marks)

[Total : 25 marks]

- Q2. (a) Recently, XYZ Company's network had been attacked by a strong malware which the perpetrator is their own staff. As a Security Manager, you want to prevent from this insider attack to happen again. Identify **FIVE (5)** methods to defense against insider attack. (8 marks)

- (b) Virus has **FOUR (4)** phases. Discuss the **FOUR (4)** phases of virus. (8 marks)

- (c) Norul works on a new project called 'ProjectX'. As a Project Team Manager, she is responsible to assign and manage group members for her project. She has created a file named as 'ProjectX' in her Unix system.

- (i) Assume that Norul has full access on the ProjectX file. Other than herself, no one else can do anything on the file. Show the permission for the file 'ProjectX' in the format of both 10-character and octal notation. (3 marks)

- (ii) Norul wants to change the permission of the file 'ProjectX' by allowing only her team members have full access including execute the file while others who are not in the same user group would not be able to perform anything to the file. Write the Unix command to set the appropriate permission on the file and show the new permission for the file 'ProjectX' in the format of both 10-10-character and octal notation. (4 marks)

- (iii) Norul wants to assign one of her group members, Nuriyah, as the new owner of the file 'ProjectX'. Write the Unix command to change the owner of the file 'ProjectX' to Nuriyah. (2 marks)

[Total : 25 marks]

- Q3. (a) Intrusion detection system (IDS) is designed to detect suspicious activities and generate alerts when they are detected. Describe **FOUR (4)** types of automated attack that could be detected by an IDS. (8 marks)

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Q3. (Continued)

- (b) Identify the type of alarms in IDS for the following scenarios (eg: True Negative).
- (i) Alarm sounded but there is no attack on the network. (1 mark)
 - (ii) Alarm sounded when IDS detect the denial-of-service attack on the network. (1 mark)
 - (iii) No alarm sounded when a malware attack happens on the network. (1 mark)
- (c) Differentiate between Stateful Firewall and Stateless Firewall. (4 marks)
- (d) Suggest **TWO (2)** methods to secure the organization's wireless network. (4 marks)
- (e) Access Control Lists (ACL) are widely used for mitigating network attacks and controlling network traffic. Referring to the network diagram Figure 2, configure the ACLs based on the following questions: -

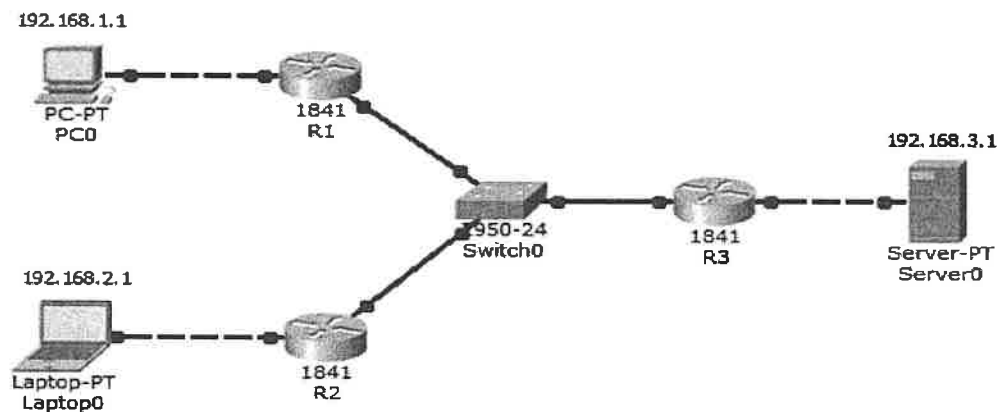


Figure 2

- (i) Write command interface to block Laptop0 from accessing to the Server0. (2 marks)
- (ii) Write command interface to allow all IP to access the Server0. (2 marks)
- (iii) Write command interface to block ICMP from accessing to the Server0. (2 marks)

[Total : 25 marks]

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Section B (Choose Any One Question)

- Q4. (a) Several cybersecurity and professional bodies were established in Malaysia. Identify **THREE (3)** professional body and their purpose. (6 marks)
- (b) Risk assessment evaluates the relative risk for each vulnerability. Assume Asset A has a value score 60 and has 1 vulnerability. Vulnerability 1 has a likelihood of 1.0 with no current controls. It is estimate that data are 93% accurate. Calculate the risk based on the given information. (5 marks)
- (c) Describe **FIVE (5)** strategies to control risk. (10 marks)
- (d) A policy is a set of ideas or a plan of what to do in particular situations that has been agreed to officially by a group of people, a business organization, a government, or a political party. Give **TWO (2)** types of policy available in information security with its description. (4 marks)
- [Total : 25 marks]

- Q5. (a) Give **THREE (3)** purposes to conduct computer forensic. (6 marks)
- (b) Describe **FOUR (4)** types of evidences in computer forensic. (8 marks)
- (c) If evidence will be used in court proceedings or actions that could be challenged legally, evidence must meet certain standards. Identify these **THREE (3)** standards of evidence. (6 marks)
- (d) Assume you as a Police Forensic Investigator investigating a homicide case. You receive a desktop computer as one of the evidence. You suspect some of the files in the computer had been deleted. Are the files really had been deleted? Justify your answer. (5 marks)
- [Total : 25 marks]

